

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Semestertoets 3
VRAESTEL

Kopiereg voorbehou

Module EBN111
Elektrisiteit en Elektronika
May 2017

Semester Test 3
QUESTION PAPER

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Module EBN111
Electricity and Electronics
May 2017



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:
Exam information:

MEMO

Maksimum punte: <i>Maximum marks:</i>	46	Volpunte: <i>Full marks:</i>	45
Duur van vraestel: <i>Duration of paper:</i>	90 min + 10 min om OKH vorm te voltooi <i>90 min + 10 min to complete OCR form</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		18 + OCR	

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die op die Optiese Karakter Herkenning (OKH) vorm ingevul word.
All questions must be answered on the Optical Character Recognition (OCR) form.
- Neem noukeurig kennis van die instruksies op die OKH vorm.
Take careful consideration of the instructions on the OCR form.
- Antwoorde moet eenhede insluit!
Answers must include units!
- Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	1. D le Roux
	2. D Ramotsoela
	3. X Ye

INSTRUCTIONS

Do step 0 before the start of the test.

Do step 1 in the first 90 minutes of the test

Do step 2 within the following 10 minutes.

STEP 0 (– 5 min to 0 min): Complete the front page of the OCR form.

STEP 1 (0 min to 90 min): Do your own calculations on the QUESTION PAPER. The question paper **will not** be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units. The PRACTICE ANSWER SHEET **will not** be taken in at the end of the test.

STEP 2 (90 min to 100 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2017EBN111S3**
- Use a **mechanical pencil** to complete the form so that you can erase errors.
- **Do not use commas!** Please make use of full stops.

Examples

01	–	1	.	4	5					m	A								
02	3	4	0	.	2	6				V									
03	1	5	.	2															
04	–	1	1	.	3	5	6			Ω									

INSTRUKSIES

Doen stap 0 voor die begin van die toets.

Doen stap 1 in die eerste 90 minute van die toets.

Doen stap 2 binne die daaropvolgende 10 minute.

STAP 0 (– 5 min to 0 min): Voltooi die voorblad van die OKH vorm.

STAP 1 (0 min to 90 min): Doen jou eie berekeninge op die VRAESTEL. Die VRAESTEL **word nie** aan die einde van die toets ingeneem nie. Skryf die antwoorde vir elke vraag op die OEFEN ANTWOORDBLAD. Maak seker om eenhede in te sluit. Die OEFEN ANTWOORDBLAD **word nie** ingeneem nie.

STAP 2 (90 min to 100 min): Dra die antwoorde vanaf die OEFEN ANTWOORDBLAD oor na die OKH vorm.

- Die “Assessment ID” is **2017EBN111S3**
- Gebruik ‘**n meganiese potlood** om antwoorde in te vul sodat jy foute kan uitvee.
- **Moenie kommas gebruik nie!** Maak eerde gebruik van punte.

Voorbeelde

01	–	1	.	4	5					m	A								
02	3	4	0	.	2	6				V									
03	1	5	.	2															
04	–	1	1	.	3	5	6			Ω									

SECTION / AFDELING 1 [4]

Question 01-02 [4]		Vraag 01-02 [4]	
Find V_O and V_S if $V_+ = 12$ V.		Bepaal V_O en V_S if $V_+ = 12$ V.	
01	V_O	01	V_O
02	V_S	02	V_S

$$V_+ = V_S \cdot \frac{120}{40+120} = 12 \Rightarrow V_S = 16 \text{ V}$$

$$\begin{aligned}
 V_O &= \frac{R_2 \left(1 + \frac{R_1}{R_2}\right)}{R_1 \left(1 + \frac{R_3}{R_4}\right)} V_S - \frac{R_2}{R_1} \cdot 8 \\
 &= \frac{20 \left(1 + \frac{5}{20}\right)}{5 \left(1 + \frac{40}{120}\right)} \times 16 - \frac{20}{5} \times 8 \\
 &= \frac{25 \times 16}{5 + \frac{5}{3}} - 32 = 28 \text{ V}
 \end{aligned}$$

01: $V_O = 28 \text{ V}$
➔

02: $V_S = 16 \text{ V}$
➔

SECTION / AFDELING 2 [8]

Question 03-04 [4]		Vraag 03-04 [4]	
a) Determine energy E_L stored in the inductor. b) Determine energy E_C stored in the capacitor. (The circuit is in steady-state (DC).)		a) Bepaal die energie E_L gestoor in die induktor. b) Bepaal die energie E_C gestoor in die kapasitor. (Die kringbaan is in gestadigde toestand (GS).)	
03	E_L	03	E_L
04	E_C	04	E_C

$$i_L = \frac{10V}{5k\Omega} = 2 \times 10^{-3} A$$

$$E_L = \frac{1}{2} L (i_L)^2 = \frac{1}{2} \times 10 \times 10^{-3} \times 4 \times 10^{-6} J = 0.00002 mJ$$

$$V_C = 4i_L \cdot 2k\Omega = 4 \times 2 \times 10^{-3} \times 2 \times 10^3 V = 16V$$

$$E_C = \frac{1}{2} C V^2 = \frac{1}{2} \times 2 \times 10^{-6} \times 16 \times 16 = 0.256 mJ$$

$$\underline{03: E_L = 0.00002 mJ = 20 nJ}$$

$$\underline{04: E_C = 0.256 mJ = 256 \mu J}$$

Question 05-06 [2]		Vraag 05-06 [2]	
If $v(t) = e^{-t}$ V, with $i_C(0) = 0$ A and $i_L(0) = 2$ A, determine $i(t)$:		Indien $v(t) = e^{-t}$ V, met $i_C(0) = 0$ A en $i_L(0) = 2$ A, bepaal $i(t)$:	
$i(t) = ae^{-t} + b$		$i(t) = ae^{-t} + b$	
05	a	05	a
06	b	06	b

The diagram shows a circuit with a voltage source $v(t) = e^{-t} V$ on the left. This source is connected in series with a parallel combination of two components. The first component is an inductor with an inductance of $0.5 H$. The second component is a capacitor with a capacitance of $0.5 F$. The current flowing out of the positive terminal of the voltage source is labeled $i(t)$ with an arrow pointing to the right. The current flowing through the inductor is labeled $i_L(t)$ with a downward arrow. The current flowing through the capacitor is labeled $i_C(t)$ with a downward arrow.

$$i_C(t) = C \frac{dv(t)}{dt} = 0.5 \frac{d}{dt}(e^{-t}) = -0.5e^{-t}$$

$$i_L(t) = \frac{1}{L} \int_0^t e^{-t} dt + i_L(0)$$

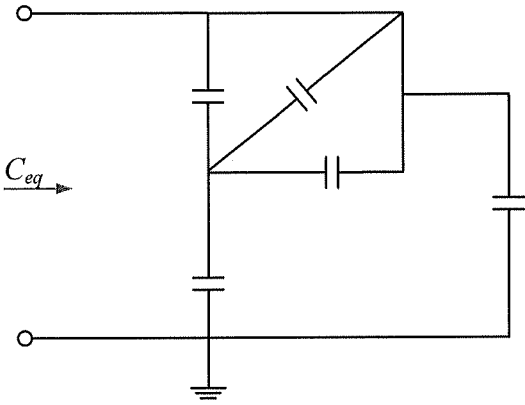
$$= \frac{1}{0.5} \int_0^t e^{-t} dt + 2$$

$$= 2(-e^{-t})_0^t + 2 = -2e^{-t} + 2 + 2 = -2e^{-t} + 4$$

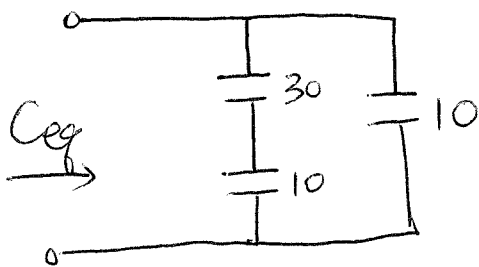
$$i(t) = i_C(t) + i_L(t) = -0.5e^{-t} - 2e^{-t} + 4 = -2.5e^{-t} + 4$$

05: $a = -2.5$ $\rightarrow$

06: $b = 4$ $\rightarrow$

Question 07 [2]		Vraag 07 [2]	
Calculate C_{eq} if all capacitors are $10\ \mu F$.		Bepaal C_{eq} indien alle kapasitoren $10\ \mu F$ is.	
07	C_{eq}	07	C_{eq}
			

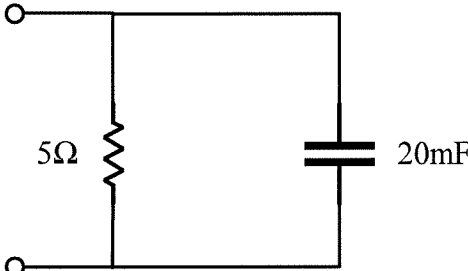
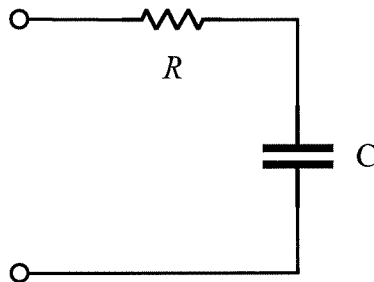
Simplify:



$$C_{eq} = 30 \parallel 10 + 10 = \frac{300}{40} + 10 = 17.5\ \mu F$$

07: $C_{eq} = 17.5\ \mu F$ $\rightarrow$

SECTION / AFDELING 3 [18]

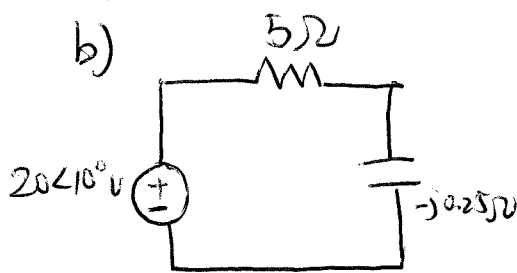
Question 08-11 [4]		Vraag 08-11 [4]	
a) Determine R and C if the input impedances of the two circuits are exactly the same at $\omega = 50 \text{ rad/s}$.		a) Bepaal R en C vir onderstaande kringbane indien dit intree impedansie van die twee kringbane dieselfde is by $\omega = 50 \text{ rad/s}$.	
08	R	08	R
09	C	09	C
			
b) If a source of $20 \cos(2000t + 10^\circ) \text{ V}$ is connected to the series combination of a 5Ω resistor and a 2 mF capacitor, what is the current that exits the positive terminal of the source? Give the final answer in the form $I = A \angle \phi$, where $A > 0$ and $-180^\circ \leq \phi \leq 180^\circ$.		b) Indien 'n bron van $20 \cos(2000t + 10^\circ) \text{ V}$ aan die serie kombinasie van 'n 5Ω resistor en 'n 2 mF kapasitor gekoppel word, wat is die stroom wat uit die positiewe terminaal van die bron vloei? Gee die finale antwoord in die vorm $I = A \angle \phi$, met $A > 0$ en $-180^\circ \leq \phi \leq 180^\circ$.	
10	A	10	A
11	ϕ	11	ϕ

$$a) Z_1 = 5 \parallel (-j \frac{1}{50 \times 20 \times 10^{-3}}) = 5 \parallel (-j) = \frac{-j5}{5-j} = 0.19230 - j0.96154 (\Omega)$$

$$Z_2 = R - j \frac{1}{\omega C}, \text{ if } Z_1 = Z_2, \text{ then } R = 0.19230 \Omega = 192.30 \text{ m}\Omega$$

$$\frac{1}{\omega C} = 0.96154 \Rightarrow C = 20.800 \text{ mF}$$

$$08: R = 192.30 \text{ m}\Omega, \quad 09: C = 20.800 \text{ mF}$$



$$Z_C = -j \frac{1}{\omega C} = -j \frac{1}{2000 \times 2 \times 10^{-3}} = -j0.25 \Omega$$

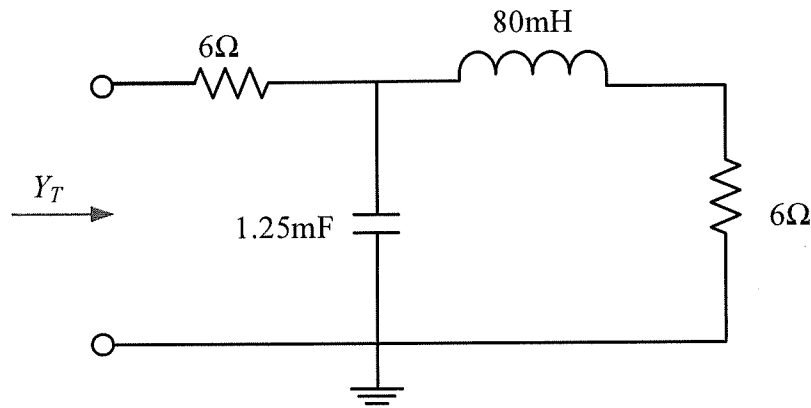
$$Z = 5 - j0.25 (\Omega)$$

$$\bar{I} = \frac{20 \angle 10^\circ}{5 - j0.25} = 3.9950 \angle 12.862^\circ (\text{A})$$

$$10: A = 3.9950 (\text{A})$$

$$11: \phi = 12.8624^\circ$$

Question 12-13 [2]		Vraag 12-13 [2]	
Find the input admittance Y_T at $\omega = 100 \text{ rad/s}$. Give the final answer in rectangular form : $Y_T = Y_a + jY_b \text{ S}$ Hint: Admittance is inverse of impedance: $Y = 1/Z$ [Siemens].		Bepaal die intree admitansie Y_T by $\omega = 100 \text{ rad/s}$. Gee die finale antwoord in die reghoekige vorm : $Y_T = Y_a + jY_b \text{ S}$ Wenk: admitansie is die omgekeerde van impedansie: $Y = 1/Z$ [Siemens].	
12	Y_a	12	Y_a
13	Y_b	13	Y_b



$$Z_C = -j \frac{1}{1.25 \times 10^{-3} \times 100} = -j8 \Omega$$

$$Z_L = j\omega L = j \times 100 \times 80 \times 10^{-3} = j8 \Omega$$

$$Z_T = 6 + (-j8) \parallel (6 + j8) = 6 + \frac{-j8(6 + j8)}{6}$$

$$= \frac{36 + 64 - j48}{6} = \frac{100 - j48}{6}$$

$$Y_T = \frac{1}{Z_T} = \frac{6}{100 - j48} = 0.048765 + j0.023407 \text{ (S)}$$

$$12: Y_a = 0.048765 \text{ S} = 48.765 \text{ mS}$$

→

$$13: Y_b = 0.023407 \text{ S} = 23.407 \text{ mS}$$

→

Question 14-17 [4]		Vraag 14-17 [4]	
a) Simplify the following expression of $\bar{V}$ and write the answers in rectangular form . $a + jb = \frac{(6 - j8)^2}{\sqrt[3]{27 \angle 135^\circ} + j10 \cos(60^\circ)}$		a) Vereenvoudig die volgende uitdrukking en skryf die antwoord in reghoekige vorm . $a + jb = \frac{(6 - j8)^2}{\sqrt[3]{27 \angle 135^\circ} + j10 \cos(60^\circ)}$	
b) Determine $\bar{V}_2$ in polar form ($A \angle \theta$). $\begin{bmatrix} 3 + j4 & -j2 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} \bar{V}_1 \\ \bar{V}_2 \end{bmatrix} = \begin{bmatrix} 3 \angle 0^\circ \\ 0 \end{bmatrix}$		b) Bepaal $\bar{V}_2$ in polêre vorm ($A \angle \theta$). $\begin{bmatrix} 3 + j4 & -j2 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} \bar{V}_1 \\ \bar{V}_2 \end{bmatrix} = \begin{bmatrix} 3 \angle 0^\circ \\ 0 \end{bmatrix}$	
14	a	14	a
15	b	15	b
16	A	16	A
17	θ	17	θ

$$a) \quad a + jb = \frac{-28 - j96}{3 \angle 45^\circ + j5} = -13.458 - j0.076970$$

$$14: a = -13.458 \rightarrow$$

$$15: b = -0.076970 \rightarrow$$

$$b) \quad V_2 = \frac{\begin{vmatrix} 3+j4 & 3 \\ 2 & 0 \end{vmatrix}}{\begin{vmatrix} 3+j4 & -j2 \\ 2 & 5 \end{vmatrix}} = \frac{-6}{15+j20+j4} = \frac{-6}{15+j24} \Rightarrow$$

$$\rightarrow = 0.21200 \angle 122.01^\circ$$

$$16: A = 0.21200 \rightarrow$$

$$17: \theta = 122.01^\circ \rightarrow$$

Questions 18-21 [4]		Vraag 18-21 [4]	
Given two sinusoid signals v_1 and v_2 , where $v_1 = 25 \cos(15t - 30^\circ)$, $v_2 = -50 \sin(15t + 150^\circ)$.		Twee sinusiedseine v_1 en v_2 word gegee: $v_1 = 25 \cos(15t - 30^\circ)$, $v_2 = -50 \sin(15t + 150^\circ)$.	
a) Determine $v = v_1 + v_2$, and write the final answer in the form $B \cos(\omega t + \phi)$ where $B > 0$ and $-180^\circ \leq \phi \leq 180^\circ$.		a) Bepaal $v = v_1 + v_2$, en skryf die finale antwoord in die vorm $B \cos(\omega t + \phi)$ met $B > 0$ en $-180^\circ \leq \phi \leq 180^\circ$.	
b) Which statement is correct? 1. v_1 leads v_2 2. v_2 leads v_1		b) Watter stelling is korrek? 2. v_1 loop voor v_2 2. v_2 loop voor v_1	
c) What is the phase angle ϕ_d ($0^\circ \leq \phi_d \leq 180^\circ$) between v_1 and v_2 ?		c) Wat is die fasehoek ϕ_d ($0^\circ \leq \phi_d \leq 180^\circ$) tussen v_1 en v_2 ?	
18	B	18	B
19	ϕ	19	ϕ
20	Choose 1 or 2.	20	Kies 1 of 2.
21	ϕ_d	21	ϕ_d

$$a) \quad v_1 = 25 \cos(15t - 30^\circ) \Rightarrow \bar{V}_1 = 25 \angle -30^\circ$$

$$v_2 = -50 \sin(15t + 150^\circ) \Rightarrow \bar{V}_2 = 50 \angle -120^\circ$$

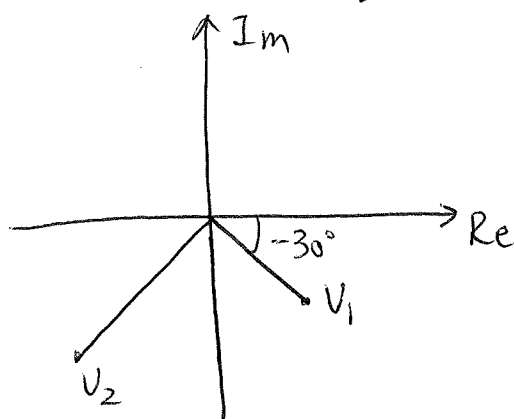
$$= 50 \cos(15t - 120^\circ)$$

$$\bar{V} = \bar{V}_1 + \bar{V}_2 = 25 \angle -30^\circ + 50 \angle -120^\circ = 55.902 \angle -93.435^\circ$$

$$18: B = 55.902 \rightarrow$$

$$19: \phi = -93.435^\circ \rightarrow$$

b)



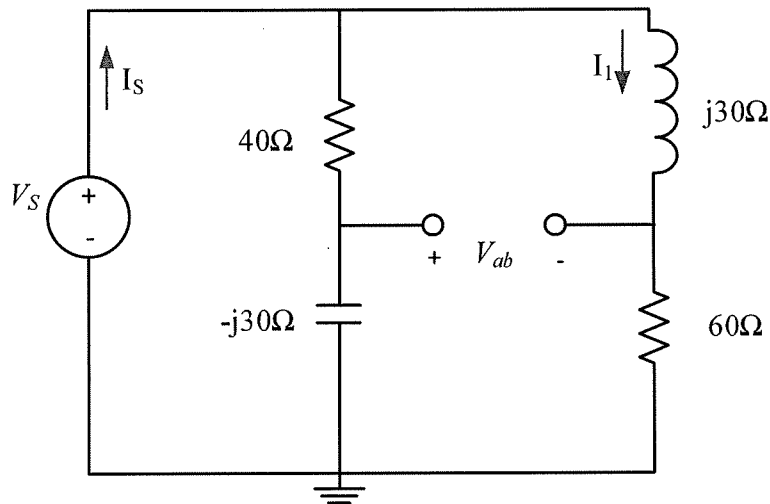
1: v_1 leads v_2

$$20: 1 \rightarrow$$

$$c) \quad \phi_d = [30^\circ - (-120^\circ)] = 90^\circ$$

$$21: \phi_d = 90^\circ \rightarrow$$

Question 22-25 [4]		Vraag 22-25 [4]	
a) Find $\bar{V}_{ab} = V_c + jV_d$ V if $\bar{V}_s = 100\angle 30^\circ$ V.		a) Bepaal $\bar{V}_{ab} = V_c + jV_d$ V indien $\bar{V}_s = 100\angle 30^\circ$ V.	
b) Find $\bar{I}_1 = I_c + jI_d$ A if $\bar{I}_s = 10\angle 0^\circ$ A.		b) Bepaal $\bar{I}_1 = I_c + jI_d$ A indien $\bar{I}_s = 10\angle 0^\circ$ A.	
22	V_c	22	V_c
23	V_d	23	V_d
24	I_c	24	I_c
25	I_d	25	I_d



$$\begin{aligned}
 V_{ab} &= V_a - V_b = 100\angle 30^\circ \cdot \left(\frac{-j30}{40-j30} - \frac{60}{60+j30} \right) \\
 &= 100\angle 30^\circ \cdot \frac{-j30(60+j30) - 60(40-j30)}{(40-j30)(60+j30)} \\
 &= 100\angle 30^\circ \cdot \frac{-15}{33-j6} = -34.105 - j28.928
 \end{aligned}$$

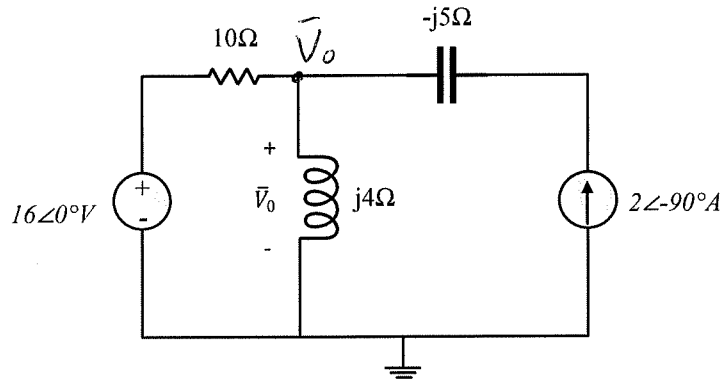
$$\underline{22: V_c = -34.105 \text{ V}}, \quad \underline{23: V_d = -28.928 \text{ V}}$$

$$\bar{I}_1 = 10\angle 0^\circ \cdot \frac{40-j30}{100} = 4-j3 \text{ (A)}$$

$$\underline{24: I_c = 4 \text{ A}}, \quad \underline{25: I_d = -3 \text{ (A)}}$$

SECTION / AFDELING 4 [16]

Question 26-27 [2]		Vraag 26-27 [2]	
Determine $\bar{V}_0$ in polar form ($C\angle\psi$) V.		Bepaal $\bar{V}_0$ in polêre vorm ($C\angle\psi$) V.	
26	C	26	C
27	ψ	27	ψ



$$2\angle-90^\circ = \frac{\bar{V}_0 - 16\angle 0}{10} + \frac{\bar{V}_0}{j4}$$

$$\Rightarrow -j2 = \frac{\bar{V}_0 - 16}{10} + \frac{\bar{V}_0}{j4} \quad (1)$$

$$(1) \times 20 \Rightarrow -j40 = 2\bar{V}_0 - 32 - j5\bar{V}_0$$

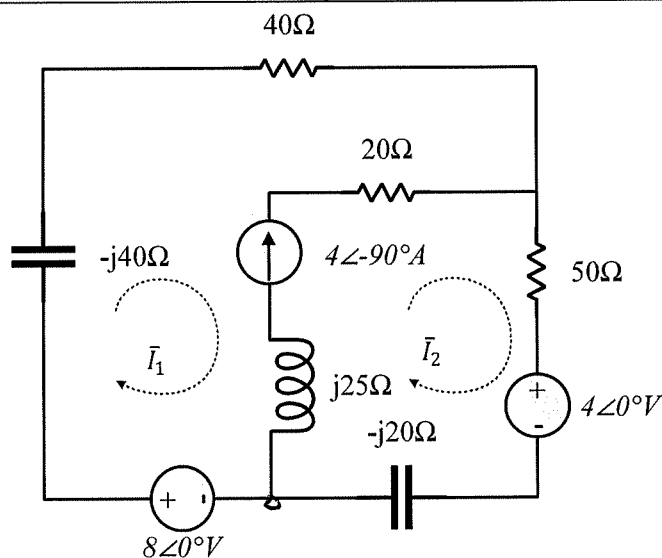
$$(2 - j5)\bar{V}_0 = 32 - j40$$

$$\Rightarrow \bar{V}_0 = \frac{32 - j40}{2 - j5} = 9.5122\angle 16.858^\circ \text{ V}$$

$$26: C = 9.5122 \text{ V}$$

$$27: \psi = 16.858^\circ$$

Question 28-33 [4]		Vraag 28-33 [4]	
Use mesh analysis to set up two equations of the following form: $\begin{cases} \bar{I}_1 - \bar{I}_2 = A \angle \theta^\circ \\ a\bar{I}_1 + b\bar{I}_2 = 4 \angle 0^\circ \end{cases}$ Give $a = x_1 + jy_1$ and $b = x_2 + jy_2$ in rectangular form , and $A > 0$, $-180^\circ \leq \theta \leq 180^\circ$.		Gebruik lus analyse en formuleer twee vergelykings in die volgende vorm: $\begin{cases} \bar{I}_1 - \bar{I}_2 = A \angle \theta^\circ \\ a\bar{I}_1 + b\bar{I}_2 = 4 \angle 0^\circ \end{cases}$ Gee $a = x_1 + jy_1$ en $b = x_2 + jy_2$ in reghoekige vorm , en $A > 0$, $-180^\circ \leq \theta \leq 180^\circ$.	
28	A	28	A
29	θ	29	θ
30	x_1	30	x_1
31	y_1	31	y_1
32	x_2	32	x_2
33	y_2	33	y_2



Mesh 1: $\bar{I}_2 - \bar{I}_1 = 4 \angle -90^\circ \Rightarrow -\bar{I}_1 + \bar{I}_2 = 4 \angle -90^\circ$
 $\bar{I}_1 - \bar{I}_2 = -4 \angle -90^\circ = 4 \angle 90^\circ$

Super mesh: $-8 \angle 0^\circ + (-j40)\bar{I}_1 + 40\bar{I}_1 + 50\bar{I}_2 + 4 \angle 0^\circ + (-j20)\bar{I}_2 = 0$
 $(40 - j40)\bar{I}_1 + (50 - j20)\bar{I}_2 = 4 \angle 0^\circ$

$\Rightarrow 28: A = 4, \quad 29: \theta = 90^\circ$
 $a = 40 - j40 \Rightarrow 30: x_1 = 40 \quad 31: y_1 = -40$

$b = 50 - j20 \Rightarrow 32: x_2 = 50$
 $33: y_2 = -20$

Question 34-37 [4]		Vraag 34-37 [4]	
Find the Norton equivalent with respect to terminals a - b . Write the answers in rectangular form , where $\bar{I}_N = I_x + jI_y$ A and $Z_N = Z_x + jZ_y \Omega$.		Bepaal die Norton ekwivalent met verwysing tot terminale a - b . Skryf die antwoorde in reghoekige vorm , met $\bar{I}_N = I_x + jI_y$ A en $Z_N = Z_x + jZ_y \Omega$.	
34	I_x	34	I_x
35	I_y	35	I_y
36	Z_x	36	Z_x
37	Z_y	37	Z_y

$$Z_N = (-j50) \parallel (80 + j100)$$

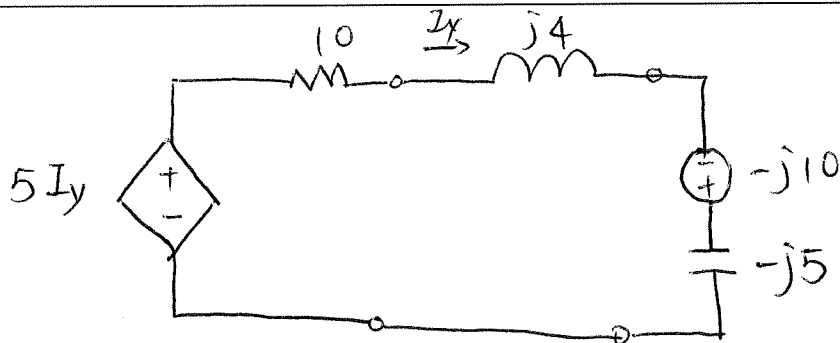
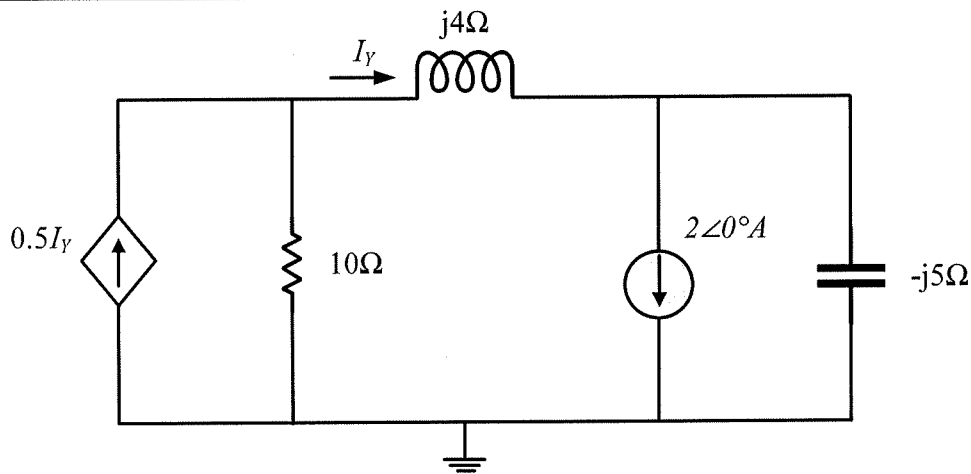
$$= \frac{-j50(80 + j100)}{80 + j50} = 22.472 - j64.045 (\Omega)$$

$$I_N = \frac{16\angle 0^\circ}{-j50} = j0.32 \text{ (A)}$$

$$\Rightarrow \underline{34: I_x = 0 \text{ A}} \quad \underline{35: I_y = 0.32 \text{ A}}$$

$$\underline{36: Z_x = 22.472 \Omega} \quad \underline{37: Z_y = -64.045 \Omega}$$

Question 38-39 [2]		Vraag 38-39 [2]	
Use source transformation to determine $\bar{I}_Y$ in rectangular form ($I_a + jI_b$) A.		Gebruik brontransformatie om $\bar{I}_Y$ te bepaal in rechthoekige vorm ($I_a + jI_b$) A.	
38	I_a	38	I_a
39	I_b	39	I_b



Use source transformation :

$$(5\bar{I}_Y - j10) = \bar{I}_Y (10 + j4 - j5) = 10\bar{I}_Y - j\bar{I}_Y$$

$$\Rightarrow \bar{I}_Y = \frac{-j10}{5-j} = 0.38462 - j1.9231$$

$$\Rightarrow^{38} I_a = 0.38462$$

$$39: I_b = -1.9231$$

Question 40-43 [4]		Vraag 40-43 [4]	
a) Determine $\bar{I}_L = I_{(A)a} + jI_{(A)b}$ A only due to the current source.		a) Bepaal $I_{(A)a} + jI_{(A)b}$ A slegs as gevolg van die stroombron.	
b) Determine $\bar{I}_L = I_{(V)a} + jI_{(V)b}$ A only due to the voltage source.		b) Bepaal $\bar{I}_L = I_{(V)a} + jI_{(V)b}$ A slegs as gevolg van die spanningsbron.	
40	$I_{(A)a}$	40	$I_{(A)a}$
41	$I_{(A)b}$	41	$I_{(A)b}$
42	$I_{(V)a}$	42	$I_{(V)a}$
43	$I_{(V)b}$	43	$I_{(V)b}$

a) Let voltage source 0, then use current division

$$I_L^{(A)} = -80\angle 0^\circ \cdot \frac{30}{30 + j40} = -28.8 + j38.4 \text{ (A)}$$

b) Let current source 0, then

$$I_L^{(V)} = 200\angle -90^\circ / (30 + j40) = -3.2 - j2.4 \text{ (A)}$$

$$\Rightarrow \begin{array}{ll} \underline{40: I_{(A)a} = -28.8 \text{ A}} & \underline{41: I_{(A)b} = 38.4 \text{ A}} \\ \underline{42: I_{(V)a} = -3.2 \text{ A}} & \underline{43: I_{(V)b} = -2.4 \text{ A}} \end{array}$$

PRACTICE PAGE / OEFENBLAD

Student number

Assessment ID

2	0	1	7	E	B	N	1	1	1	S	3		

Answer

Unit

01	$V_O = 28$	V
02	$V_S = 16$	V
03	$E_L = 0.00002$	$mJ = 20 \text{ nJ}$
04	$E_C = 0.256$	$mJ = 256 \text{ uJ}$
05	$a = -2.5$	
06	$b = 4$	
07	$C_{eq} = 17.5$	μF
08	$R = 192.30$	$m\Omega$
09	$C = 20.800$	mF
10	$A = 3.9950$	A
11	$\phi = 12.862$	$^\circ$
12	$Y_a = 0.048765$	$S = 48.765 \text{ mS}$
13	$Y_b = 0.023407$	$S = 23.407 \text{ mS}$
14	$a = -13.458$	
15	$b = -0.076970$	
16	$A = 0.21200$	
17	$\theta = 122.01$	$^\circ$
18	$B = 55.902$	
19	$\phi = -93.435$	$^\circ$
20	1	
21	$\phi_a = 90^\circ$	$^\circ$
22	$V_c = -34.105$	V
23	$V_d = -28.928$	V
24	$I_c = 4$	A
25	$I_d = -3$	A

26	$C = 9.5122$	V
27	$\psi = 16.858$	$^{\circ}$
28	$A = 4$	
29	$\theta = 90$	$^{\circ}$
30	$x_1 = 40$	
31	$y_1 = -40$	
32	$x_2 = 50$	
33	$y_2 = -20$	
34	$I_x = 0$	A
35	$I_y = 0.32$	A
36	$Z_x = 22.472$	Ω
37	$Z_y = -64.045$	Ω
38	$I_a = 0.38462$	A
39	$I_b = -1.9231$	A
40	$I_{(A)a} = -28.8$	A
41	$I_{(A)b} = 38.4$	A
42	$I_{(V)a} = -3.2$	A
43	$I_{(V)b} = -2.4$	A